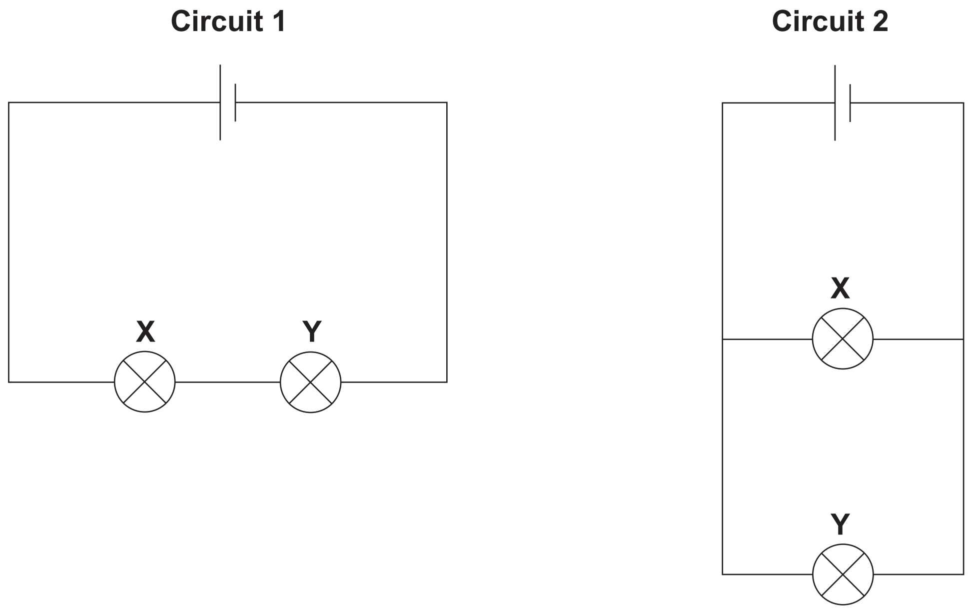
Examiner only

**2.** A student is studying the properties of different electrical circuits. **X** and **Y** are identical lamps.

The student built the following circuits.




*(a)* Complete the sentences using the terms in the box.
*The terms can be used once, more than once or not at all.*

| | | | |
|---|---|---|---|
| series | less than | parallel | the same as |
| greater than | simple | current | voltage |

(i) The lamps in **circuit 1** are connected in .................................................... .

The current through lamp **X** is ....................................................the current through lamp **Y**. [2]

(ii) The lamps in **circuit 2** are connected in .................................................... .

The current through lamp **X** is ....................................................the current through lamp **Y**. [2]

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*(b)* In **circuit 1** the voltage supplied is 9 V. Each lamp has a resistance of 3 Ω.

(i) Use the equation:

$$\text{current} = \frac{\text{voltage}}{\text{resistance}}$$

to calculate the current in this circuit. [3]

Current = .............................. A

(ii) State what would happen to the current in **circuit 1** if lamp **Y** is removed and the circuit was built with only lamp **X**. [1]

..............................................................................................................................................................

*(c)* In **circuit 2** the voltage supplied is also 9 V and each bulb has a current of 3 A.

Use the equation

$$\text{power} = \text{voltage} \times \text{current}$$

to calculate the power of lamp **X**. [2]

Power = .............................. W

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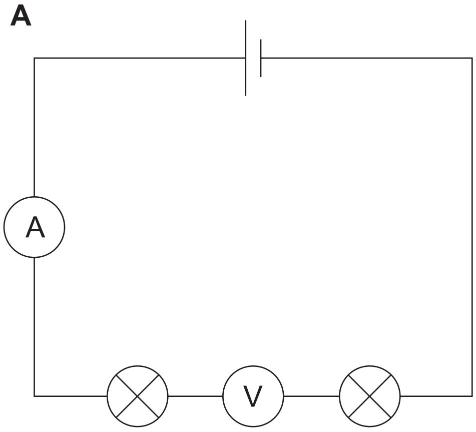
*(d)* The student set up a circuit to measure the current and voltage correctly.

Use a tick (✓) to identify the circuit diagram that should have been used to measure the voltage. [1]

**A**

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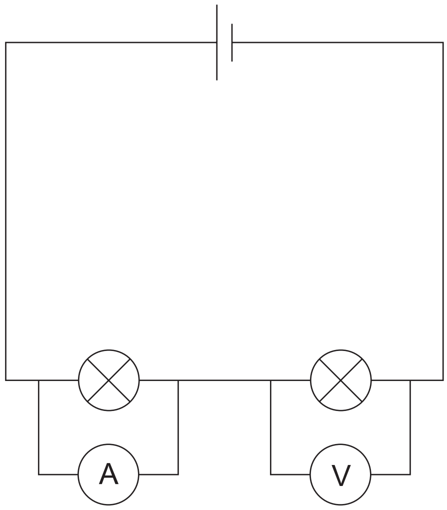
**B**

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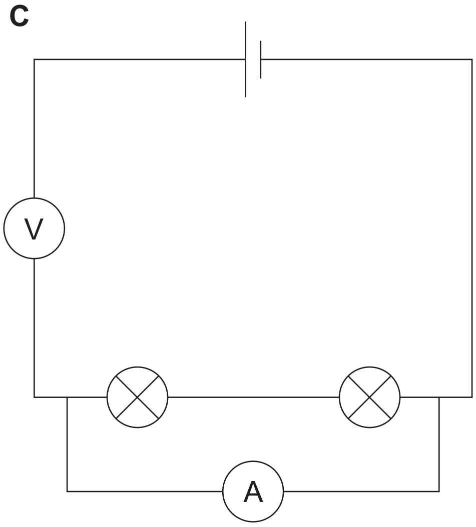
**C**

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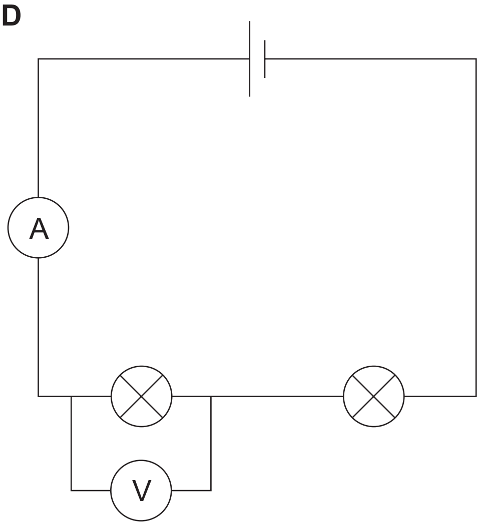
**D**

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*(e)* State **two** reasons why lamps are connected in parallel rather than in series in the home. [2]

1. ..........................................................................................................................................

2. ..........................................................................................................................................

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